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Invited Review

A review of operational spare parts service logistics in service control towers

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ABSTRACT

In this paper, we provide an overview of operational spare parts planning in service control towers. We conduct company surveys to identify the issues, the challenges, and the needs in practice. We propose a classification framework to reveal the key aspects of operational planning decisions, and to provide an overview of the scientific literature. We identify promising research directions. These include incorporating service level agreements in operational decision making, managing exception messages, integrating tactical and operational plans, using advance supply and demand information for operational interventions, and developing a unified and holistic approach for selecting interventions in large scale systems.
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1. Introduction

Capital goods are advanced technical systems that are important for the delivery of goods and services. Examples are manufacturing machines, advanced medical equipment, industrial computers, material handling systems, and defense systems. The availability of such systems is crucial for guaranteeing uninterrupted delivery of these vital goods and services. Operators of these systems typically aim to reduce life cycle costs and to achieve uptime. These goals require specialized knowledge of these systems. Therefore, after sales services are often provided by original equipment manufacturers (OEMs) and third-party service providers to the users. These services range from spare parts supply to full maintenance services covering preventive and corrective maintenance.

Service commitments to customers are defined in service contracts known as *service level agreements* (SLAs). SLAs include penalty clauses that apply if service requirements are not attained. In some SLAs, which are known as performance-based contracts, the operators do not own the systems but pay a flat fee per hour system usage (Kim, Cohen, & Netessine, 2007). Therefore, service levels are high, e.g., the mean waiting time for spare parts is expressed in hours, and tracking service levels during the contract period is crucial to avoid penalties. Service providers look for a balance between achieving high service levels by holding spare parts close to customer sites and limiting inventory and expensive solutions such as emergency shipments.

To achieve these goals, next to tactical and strategical level planning decisions, companies need to track day-to-day performance. They intervene when a stockout occurs, or preferably prevent that a stockout will occur. To perform these actions, planners may benefit from real supply and demand time information. The new type of information and decision support systems that merge and visualize this supply chain information are called *supply chain control towers* (Bhosle et al., 2011; Rustenburg, 2016). As the concept is relatively new the definition and capabilities are still evolving in time. For after-sales service supply chains, we will use the term Service Control Towers (SCTs).

A service control tower acts as a centralized hub that uses real-time data from a company's existing, integrated data management and transactional systems to integrate processes and tools across the end-to-end supply service chain and drives business outcomes (Accenture, 2015). Companies (aim to) use SCTs to monitor their supply chain and to generate alerts, also known exception messages or review reasons. These messages are typically generated based on business rules. Planners review these messages and select an operational intervention, such as placing an emergency shipment or an extra new buy. Due to the considerable number of such messages, planners can only process a fraction of these

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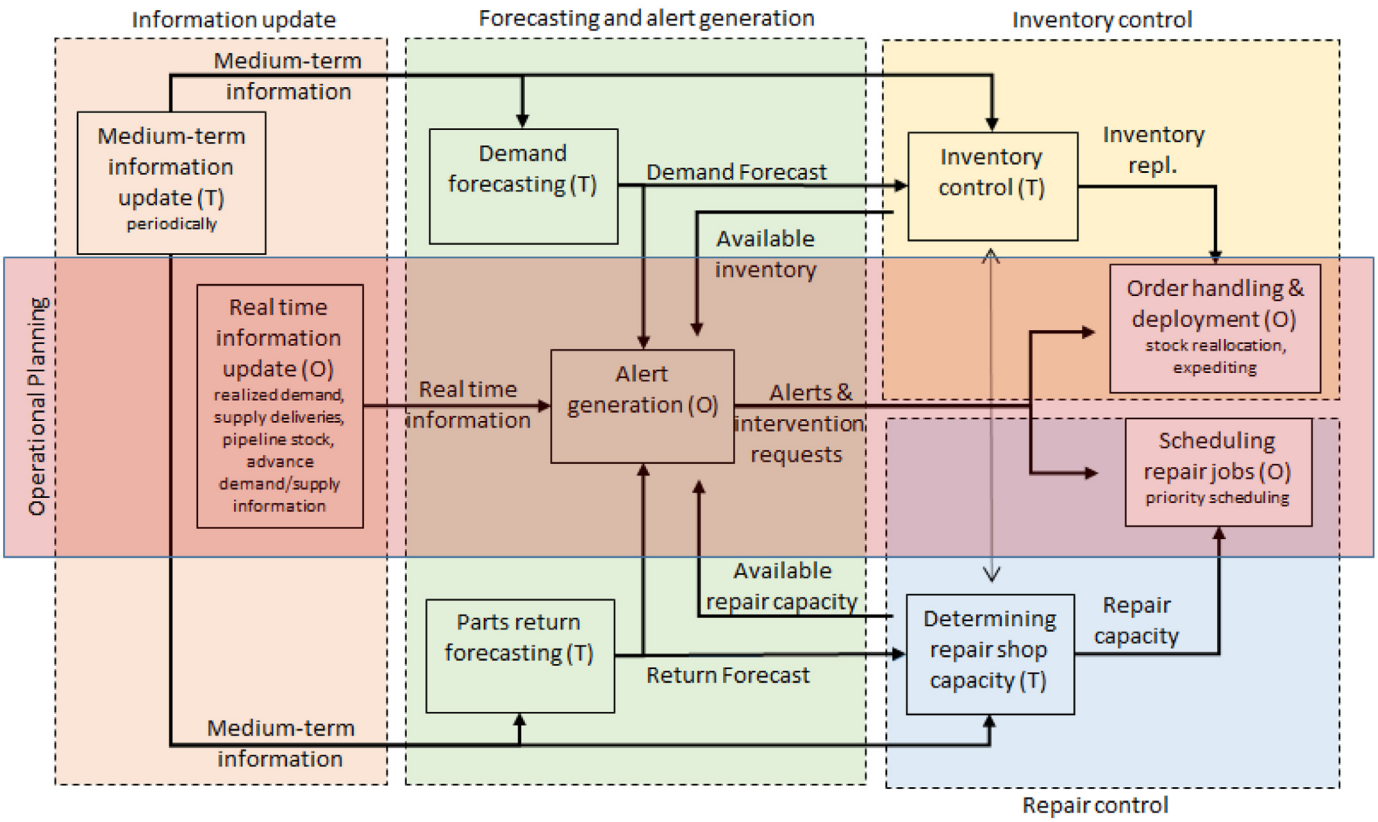

Fig. 1. Tactical and operational decisions and processes.

messages. Therefore, it is essential to prioritize messages and to automate standard operational interventions. Yet, the SCTs that we are aware of lack analytics functionality to properly assess the impact of interventions and support the day-to-day operational planning decisions although they are well equipped to provide real-time information. On the other hand, there are several papers on operational planning in the literature, which may help in supporting these decisions.

A typical SCT consists of five layers: supply chain business, data perception, data storage and control, data application, and operational planning (Shou-Wen, Ying, & Yang-Hua, 2013, here we use slightly different names than the paper). We are particularly interested in the operational planning layer. The data application and supply chain business layers are partly important for our review because the first addresses metrics, alerts and demand trends and the second the supply information needed (supply characteristics) in making operational decisions. We discuss improvement potentials for the data application layer in Section 4.2.3. The other layers are outside our scope, because decisions regarding these layers relate to tactical or strategical planning, or these layers belong to other research areas such as information systems.

The day-to-day operational planning decisions in practice are characterized by the following features:

- The decisions have immediate effect or take a shorter time than e.g. regular replenishments; they mostly influence short-term performance (hours, days, or at most weeks).
- Decisions are based on real-time information, such as current inventory on hand, pipeline stock, number of parts in repair or return, process completion time estimates, short-term demand forecasts and advance demand information (ADI) derived from condition monitoring or preventive maintenance plans. Such information is not available for tactical planning.
- Strategic and tactical decisions, e.g., supply network, inventory control parameters, cannot be influenced in short-term, and therefore they are fixed.

Fig. 1 portrays the hierarchical relation between tactical (indicated by T) and operational (indicated by O) decisions in maintenance and spare parts planning, and also their relation to information use. Different colors show content-wise categorization of decisions, e.g., inventory control, repair control. In this review, we focus on operational level decisions (middle horizontal block in Fig. 1). For a detailed discussion on hierarchical relations in maintenance and spare parts planning, we refer to Driessen, Arts, van Houtum, Rustenburg, & Huisman, 2015.

Our review contributes to the spare parts supply management literature, which is represented by the following books: Muckstadt, 2005; Sherbrooke, 2004; Van Houtum & Kranenburg, 2015. Several frameworks have been developed, each focusing on different levels of spare parts planning (Cavalieri, Garetti, Macchi, & Pinto, 2008, on tactical planning; Driessen et al. (2015), on hierarchical planning of tactical and operational decisions; Hu, Boylan, Chen, & Labib, 2018, on lifecycle of parts; Huiskonen, 2001, on strategical and tactical planning). There is also a benchmark analysis focusing on strategical and tactical planning (Cohen, Zheng, & Agrawal, 1997). Also, several reviews are conducted on different aspects of spare parts planning, e.g., age-based maintenance, multi-echelon problems, obsolescence, repairable (Kennedy, Wayne Patterson, & Fredendall, 2002); multi-item models, service differentiation, lateral transshipments (Van Houtum, Rustenburg, & Huisman, 2015); system-oriented spare parts inventory models (Basten & van Houtum, 2014); maintenance and service logistics of maritime assets (Eruguz, Tan, & van Houtum, 2017). We differ from these frameworks and reviews in two ways: First, we focus only on operational level planning. To the best of our knowledge, our paper

Table 1
The characteristics of the service structure of the five companies.

Service structure (current situation)	OEMs				
	A	B	C	D	E
Number of product types	Many (>1000)	Medium (100–1000)	Many (>1000)	Few (customized) (<100)	Medium (customized) (100–1000)
Number of installed bases / Number of customers	Very many (>10,000)/Many (>1000)	Many (>1000)/Few (<100)	Medium (100–1000)/Medium (100–1000)	Few (<100)/Few (<100)	Many (>1000)/Many (> 750)
Number of customers / Type of customers	Many (>1000)/B2B and B2C	Few (100–1000)/B2B	Few (100–1000)/B2B	Very few (<100)/B2G	Many (>1000)/B2B
Number of echelons / Number of warehouses	3–4 echelons/3 CWs, hundreds of LWs	2–3 echelons/1 CW, <50 LWs	2-echelon/1 CW, <20 LWs	2 echelon/1CW, 4 assets as depot	1 echelon/1CW
Carries out maintenance? / Operates repair shops?	Yes/Yes (as a separate entity)	Yes, onsite/No	No/Yes	No/Yes	Yes, onsite/No
Parts planning: LRU and or SRU / Number of SKUs	LRU/Many (>10,000)	LRU/Medium (1000-10,000)	LRU and SRU/Many (>10,000)	LRU and SRU/Medium (1000–10,000)	LRU/Medium (1000–10,000)
Part characteristics (demand and price)	All sorts of demand rates and prices	Mostly slow moving (<5 per year) and expensive (>1000€)	All sorts of demand rates and expensive (>1000€)	Mostly slow moving (<5 per year) and expensive (>1000€)	All sorts of demand rates, typically cheap (<100€)
Workforce among operational and tactical planning	Operational: 40% FTEs Tactical: 60% FTEs	Operational: 70% FTEs Tactical: 30% FTEs	Operational: 67% FTEs Tactical: 33% FTEs	Operational: 60–90% FTEs Tactical: 40–10% FTEs	Operational: 90% FTEs Tactical: 10% FTEs

is the first review paper and the first to propose a framework on the operational planning of spare parts supply. Second, we focus also on the current state-of-art in practice, and do not confine to research papers in the literature. There are reviews on specific aspects that are somewhat related to our review, i.e., lateral transshipments (Paterson, Kiesmüller, Teunter, & Glazebrook, 2011), repairable inventories (Guide & Srivastava, 1997) and inventory rationing (Teunter & Klein Haneveld, 2008). In contrast to these papers, we do not focus on only one specific subject, but we cover a broad range of operational interventions in spare parts planning. We focus on spare parts only, being aware of the fact that other resources such as service engineers and tools are also important to maintain capital goods (e.g., Hill, 1992). Most companies in this field (including all those directly inspire our paper) consider spare parts as the most crucial resource. A key reason is that spare parts may lead to significant downtime due to long replenishment lead times.

Our contribution is fourfold: (i) We identify the challenges of operational spare parts planning in SCTs, e.g., operational interventions, short term planning, short-term performance, real-time information and triggers, alert generation, and the link between tactical and operational planning while leaving the information technology aspects outside our scope. (ii) We propose a framework to identify key aspects of operational planning decisions. (iii) We provide an overview of research papers. (iv) We identify the gaps and recommend new research directions.

Our findings are based on three main sources: (i) a survey and direct discussions conducted at five companies participating in an ongoing research project on after-sales service of capital goods, in which an SCT is already in place or being developed, (ii) our experiences with 10–20 companies that are involved in several similar service logistics projects (https://www.dinalog.nl/en/themes/service-logistics for information about the corresponding projects and the companies), (iii) 60 research papers we identified, relating to the operational planning of spare parts supply.

The rest of the paper is organized as follows: In Section 2, we introduce the five companies at which we conducted our survey, and we provide an overview of their current practice. In Section 3, we develop a framework to classify operational decisions considering the needs and the opportunities in practice as well as modeling characteristics that we found in the literature. Using our framework as a basis, we explore the practice (mainly the companies at which we conducted our survey) in Section 4 and review the related literature in Section 5. In Section 6, we discuss the gaps in the literature, and identify opportunities for relevant research in this field.

2. The business environment in spare parts planning

In this section, we introduce the companies that form the basis of our observations and discuss their common key supply and demand characteristics. In this way, we portray the state-of-the-art and the current practice of spare parts in after sales service logistics.

Five companies participated in our survey. For sake of confidentiality, we abbreviate them as *A*, *B*, *C*, *D*, and *E*. We refer to these companies as *the five*. *A* is an information technology company on hardware systems and telecommunication devices. *B* is a world leading manufacturer in the semiconductor industry. *A* and *B* offer performance-based service contracts to their customers to provide maintenance and spare parts services. *C* operates in the aerospace industry and offers spare parts, technical support, and component repairs to aircraft operators. *D* produces professional electronics for defense and security applications. *E* produces large material handling systems for service and production facilities. *D* and *E*'s products are customized and project based. *A* and *B* want to expand the functionality of their SCTs. *C* and D are currently developing their SCT. *E* does not operate an SCT, but both has expressed interest in developing one.

The five have a strong market position or are market leader in their world-wide market. They function in business environments in which they have close ties with their customers during the entire life cycle of installed systems. Table 1 provides a comparison between *the five* based on (i) the number of product types, (ii) the number of installed bases and the number of customer locations, (iii) the number of customers, (iv) the number of echelons and the number of local warehouses (LWs) and central warehouses (CWs), (v) the maintenance functions, i.e., whether maintenance is carried out by the OEM and whether repairs are performed on-site or outsourced, (vi) whether parts are line replaceable units (LRUs) or shop replaceable units (SRUs), and the number of stock keeping units (SKUs), (vii) demand and price characteristics of parts, (viii) how planning activities (in full time equivalents, FTEs) are distributed between tactical and operational planning activities. We use labels for the various classes as defined in Table 1.

Table 1 shows that customer locations and installed bases are geographically dispersed all over the world for *the five*. To provide services to their customers, these companies mostly operate

Table 2
Software tools used in operational and tactical planning by the five companies.

Planning level	OEMs				
	A	B	C	D	E
Operational	Entercoms & some other tools	Own dashboard tool with Spotfire	Own tool (in development)	Oracle ERP (SCT in development)	Own dashboard (SCT to be developed)
Tactical	Servigistics	Own tool	Own tool (in development)	Oracle ERP, OPUS, SimLox	SLIM4

complex supply networks with two to four echelons accompanied by all sorts of part shipments in the supply network. *E* and partly *D* are exceptional in this regard as their own supply network consists of only a single central distribution center. In addition, depending on the service agreements, *D* employs assets that belong to its customers and they can be regarded as additional echelons. SKUs in spare parts assortments are often classified as slow moving and expensive items, particularly if we compare them with demand volumes and prices of items in other markets such as retailing. Operational planning requires significant effort (40% to 90%) of the spare parts planning workforce whose size varies between 5 and 40 FTEs at *the five*. *The five* use commercial software and/or develop their own tool solve these planning problems. Table 2 summarizes the software tools that *the five* use in operational and tactical planning. This table also shows differences in maturity of SCT tools.

Apart from *the five*, there are another 10–20 companies on which we base our observations. They are OEMs or maintenance and service providers of capital goods such as industrial printing, healthcare, advanced food processing, nanotechnology systems, and large vehicles. We have worked with them in the last 15–20 years in several research projects. In the rest of the text, we refer to them as *the others*. Our observations above are similar for *the others*.

3. The proposed framework

Our framework for characterizing operational planning in an SCT consists of two types of elements: (i) practice-driven elements that we observe in the companies and (ii) model-driven elements that are less visible to practitioners but still important to build decision tools. We identify practice-driven elements from our survey and the discussions conducted at *the five*. We discuss these elements in Section 4. We enrich these elements by our own observations at *the others*. The model-driven elements are derived from the literature. We discuss in Section 5.

3.1. Practice-driven elements

We identify four elements in operational planning practices: *key performance indicators (KPIs)*, *interventions*, *triggers for interventions (alert generation)*, and *information content*.

3.1.1. Key performance indicators

The KPIs, which are often associated with tactical planning, are defined in the SCT application layer and shown in SCT dashboards. They are the key objectives or constraints in operational planning. These are fill rate (fraction of demand satisfied immediately from stock), time-based fill rate (fraction of demand satisfied within a fixed period), downtime waiting for parts (cumulative time that the machine is down due to the unavailability of parts within a certain period), call resolution time and transportation performance. The latter is measured as the percentage of successful shipments in correct size within promised time, which is relevant when transportation is outsourced.

It is common in practice to measure KPIs for subsets of parts, customers, locations, or time. We refer to this as segregation. Companies use segregation in tactical planning. Therefore, segregation does not necessarily imply differentiation in operational planning. If segregated KPIs are considered in operational plans to differentiate service performance, we refer to this as differentiation. We identify four segregation types: (i) parts (per item or per system), (ii) customer (per customer, per group or for overall customers), (iii) geographical area (per location, per region, or for global network), (iv) time (over a fixed time interval or over a steady-state infinite horizon). In contrast to segregation, aggregation brings planning flexibility to attain a higher performance at a lower cost. An example is the system approach (aggregation on parts), which allows keeping less stock for expensive parts and more stock for cheaper parts to achieve an aggregate service target. The success of an SCT depends on the extent that these KPI's can be timely shown at the right level.

3.1.2. Interventions

We categorize interventions, which the SCT should suggest and evaluate, into two groups: (i) *reactive* interventions to speed up the recovery of a failed system upon a stockout, (ii) *proactive* interventions to reduce downtime risk. Interventions may depend on customer type if the service levels agreed are different (customer service differentiation). We identify the following groups of interventions:

(1) *Stock reallocation*, i.e., either balancing parts availability among downstream locations (e.g., *proactive stock reallocation*) or allocating stock for dynamic demand fulfillment (*backorder clearing*), both by using real-time information and deviating from the tactical inventory policy. Specific aspects include:
   (i) *After-repair stock reallocation*, e.g., dispatching parts after repair to the location where it is needed most,
   (ii) *Reallocation of returned parts*, e.g., dispatching unpacked, unused parts to the location in the forward supply chain where it is needed most,
   (iii) *Dynamic inventory rationing*, i.e., reserving stock for high priority requests.
   (iv) *Reallocation of parts reserved for preventive maintenance,* either for a corrective maintenance or to reduce downtime risk, thereby delaying a scheduled maintenance.
(2) *Expediting*, i.e., one-time shortening repair or replenishment lead times, where we distinguish two situations:
   (i) *The order has not been started yet.* Then, the options are:
      - *Emergency shipments*, i.e., using a faster transportation mode to resupply local inventories from the regular source upstream the supply chain,
      - *Lateral transshipments*, i.e., using a nearby warehouse at the same level in the supply chain,
      - *Increasing process capacity temporarily*, e.g., *outsourcing* repair or using *overtime* work in a repair shop,
      - *Non-preemptive priority scheduling of jobs*, e.g. at repair shops.
   (ii) *The order has already been started.* The options are:
      - *Rerouting,* i.e., changing the destination of a replenishment order.
      - *Expediting a regular delivery* of a part in the pipeline,

- *Preemptive priority scheduling of repair jobs,* i.e., changing the sequence of jobs at repair shops,
- *Expediting a new buy*, i.e., expediting a replenishment order from an external supplier by moving the delivery forward in time.

(3) *Cannibalization*, i.e., retrieving a part from a non-operating system (down or in preventive maintenance) to maintain a failed system.
(4) *Dynamic capacity allocation and lot sizing*, i.e., the number of parts to be repaired are determined reactively under finite repair capacity or proactively for flexible capacity.
(5) *Joint ordering* of different parts to save transportation costs (thereby delaying or advancing shipments). Example is consolidating shipments of different parts.

In some cases, it makes sense to omit an intervention and backorder a part request, e.g., if a regular replenishment is expected to arrive soon, or if the part is less critical.

3.1.3. Triggers for interventions

Interventions are triggered by alerts from the application layer of the SCT at fixed points in time (periodic review), or at certain events (continuous review). Examples of event-based triggers are stock level changes (fulfillment of a part request, arrival of a replenishment order), job completion (repair or return), information updates about expected timing of supply or demand, and wrong or defective part delivery. An alert is generated when actual demand (or supply) deviates significantly from expected, usually when a predetermined threshold is exceeded. This requires short-term demand (or supply) forecasting and setting thresholds for alert generation.

3.1.4. Information content

For operational level spare parts planning decisions, different types of real-time information is collected in the business layer and processed and stored in the data perception layer of an SCT. This is often a complex IT process as data in various formats need to be combined. At the supply side, examples are on-hand and pipeline inventory levels in each warehouse, and status information about return, resupply, and repair processes, completion times. At the demand side, examples include short-term demand forecasts, downtime incurred, condition monitoring (degradation) information of parts and components in installed systems. All this information is far more specific than what is available in tactical plans, or it may even deviate from the assumptions made in tactical planning.

3.2. Model-driven elements

Using our observations for *the five* and *the others* and the research papers that we review, we identify five key model-driven elements in operational planning models within an SCT: *relation between operational and tactical decisions*, *demand and supply characteristics*, *supply chain structure* and *number of parts*, *planning horizon*, and *optimization and evaluation methods*.

3.2.1. Relation between operational and tactical decisions

We identify three approaches. The first approach merely ignores the link between operational and tactical planning. The second approach assumes a *hierarchical* approach, in which tactical and operational level decisions are optimized *sequentially* (tactical first, operational next). Tactical decisions, e.g., base stock levels, are *fixed* and they serve as constraints for operational decisions. The drawback is that performance of the operational planning depends on the quality of tactical decisions. Also, the impact of operational planning on tactical planning is neglected. The third approach fully *integrates* operational and tactical planning by solving these decision problems *simultaneously* or by running a hierarchical model behind these decisions *iteratively*. Simultaneous approaches require complex policies such as dynamic threshold policies that use real-time information. Iterative approaches are based on simple principles such as rolling horizon and feedback loops. Therefore, they are more common in practice. In practice, we typically have a large variety of interventions, but integrated approaches incorporate only a limited number of interventions, and often in a simplified way.

3.2.2. Demand and supply characteristics

Demand and supply characteristics, which are often associated with tactical planning, play also an important role in operational planning. Spare parts are usually slow moving and characterized by uncertain demand. Therefore, demand is often modelled as a *Poisson* or *compound Poisson* process. In some cases, demand is modeled by a *Normal* distribution or a *general* (arbitrary) probability distribution. Also, demand can be predicted using condition monitoring of installed bases or maintenance plans. Then, demand is a function of *advance demand information* (ADI) or future demand signals. The supply processes, particularly internal supply processes (shipments, returns), typically show less uncertainty. Therefore, supply processes are generally assumed to be deterministic.

3.2.3. Supply chain structure and number of parts

The supply chain structure, which is associated with strategical planning influences operational planning. We characterize it by the number of echelons and the number of locations per echelon. We distinguish three categories: *multi-echelon with multiple locations*, *single echelon with multiple locations*, and *single echelon with single location*.

For the number of parts, we distinguish between *single-item* and *multi-item* models. Multi-item models can be subdivided based on the number of indenture levels, i.e., the number of levels in the product structure at which failed parts can be replaced by spares, e.g., LRUs and SRUs, the latter allowing replacement of subcomponents at repair shop level. We therefore classify models as *single-indenture* or *multi-indenture* models.

Next to inventory locations, the supply chain may include a repair shop. Such a repair shop can be modelled a *single processing stage* or as a *network of processing stages* representing a job shop. Each stage may consist of one or multiple parallel machines or servers.

3.2.4. Planning horizon

We identify three categories: (i) a *finite horizon with a single period* (single decision opportunity, e.g., until the next replenishment), (ii) a *finite horizon with multiple periods* (multiple decision opportunities over a fixed horizon), (iii) an *infinite horizon*, in which the horizon length is not specified and (as an approximation) a steady state analysis is assumed.

3.2.5. Optimization and evaluation methods

There are basically two methodological issues: *policy optimization*, i.e., finding the (near-) optimal policy parameters, and *policy evaluation*, i.e., evaluating the performance of a policy with given policy parameters. We identify two categories for policy optimization: *exact methods* and *heuristics*. Exact methods include *dynamic programming (DP)*, *linear programming (LP)*, *mixed integer programming (MIP)*, *enumeration,* and *nonlinear search* methods. These methods are usually computationally demanding. Heuristics can be (i) simple *decision rules* or *policies*, e.g., a threshold level to determine when to use an emergency shipment, or (ii) more complex analytical methods, e.g., *myopic/greedy* algorithms, *marginal analysis*, and *decomposition*. Heuristics are computationally less demanding but they do not guarantee optimality.

Table 3
Key performance indicators in use at *the five*.

Key performance indicators	OEMs				
	A	B	C	D	E
Fill rate	√	√	√		√
Time based fill rate	√	√	√		
Downtime waiting for parts		√		√	
Call resolution time	√	√			
Others	Transportation performance				

We identify mainly three categories for policy evaluation: *exact* evaluation, *approximation*, and *simulation*. Examples of exact methods are a transient or steady-state analysis using a continuous or discrete time Markov chain. An example of an approximation is a two-moment approximation for the pipeline stock distribution. Approximations are useful when exact evaluation is not tractable, however, their performance needs testing.

4. Survey design and empirical findings

In this section we discuss our survey. We highlight our findings and use it to portray *the five* according to the elements of our framework.

4.1. Survey design

To conduct our survey, we selected one senior supply chain manager from each of *the five* as respondent. We asked the five respondents to provide scores for each element of our framework. For KPI usage, we used a simple category scale with answer options *yes* or *no*. We used an open question regarding the time window for service levels being used. For interventions and information content, we used expert judgment with a three-point rating list scale. We defined the following score definitions: 1 = element is not important for the own company, 2 = element is moderately important, 3 = element is very important. For the remaining elements, we directly ask the respondents to characterize their systems according to our framework. The respondents are questioned independent of each other. Next, we sent the overview for all five companies to each respondent, asking whether they agreed or would change their mind. The latter was not the case. Our survey provides a good overview of practically relevant elements of operational planning from the perspective of *the five*. We discuss the responses of companies in Section 4.2. We confront our findings with observations made in the literature in Section 5. Some replies of *D* vary depending on service contract types. Since *E* does not have an SCT, model driven aspects are not applicable (denoted by NA) to *E*.

4.2. Empirical findings

4.2.1. Key performance indicators

Table 3 shows the KPIs used by *the five*. Table 4 presents how these KPIs are segregated at *the five*. Based on these tables, we make the following observations:

- Fill rate and time based fill rate are the most common KPIs. Surprisingly, downtime waiting for parts is only used by two of the five. Since downtime costs are high in spare parts logistics, we expected that the latter would be core to most (if not all) companies.
- Two of *the five* measure performance at individual item level only. The other three also measure performance at aggregate (system) level. This shows that the system approach is less common in operational planning than we typically see in tactical planning.

Table 4
Segregation of key performance indicators at *the five*.

Key performance indicators defined over	OEMs				
	A	B	C	D	E
Individual part level (item approach)	√	√	√	√	√
Set of parts (system approach)		√	√	√	
Per customer (performance based)	√	√	√	√	√
Per group of customers	√	√	√	√	
Over all customers	√		√	√	√
Per local warehouse		√	√		√
Average over a set of local warehouses	√	√	√		
Time window for service levels	Week, month	Year	Changes	Year	Not used

- *The five* measure performance also at customer level, particularly for performance-based contracts. We attribute this to the diverse service level requirements of customers.
- Four of *the five* measure performance over a fixed time interval (e.g., a quarter or a year). Yet, our direct discussions with *the five* confirm that interventions are not differentiated over fixed time windows. Otherwise, for example, when a real-time service level over a time window at one warehouse satisfies the promises, the inventory at this warehouse could be used until the end of that time window for other locations and customers.

4.2.2. Interventions

Table 5 summarizes the interventions at *the five*. We make the following observations:

- Emergency shipments from upstream are crucial. Lateral transshipments are not considered by all, e.g., because local stocks may be dedicated to certain customers.
- Expediting part return/repair is relevant for all, by changing priorities for most.
- Not intervening (usually backordering) is also an option, despite emphasis on emergency shipments in practice as well as in recent literature.
- Cannibalization is strongly business dependent. For example, it is common in aviation. Using parts reserved for preventive maintenance to resolve a failure is an option for most.
- *The five* all pay some attention to joint ordering of different parts to save shipment costs.
- Dynamic inventory rationing appears as the only intervention facilitating customer differentiation. Three of *the five* apply either limited or no dynamic rationing. This is surprising because customer differentiation is widely used in tactical planning (e.g., in determining base stock levels per customer location).

4.2.3. Triggers for interventions

All *five* use a mix of time- and event-driven triggers. Yet, event-driven triggers, which are often based on thresholds, are more common. Based on other survey questions and direct discussions on triggers and alert generation at *the five*, we make the following observations:

- Despite using customized software tools, companies struggle with several aspects of data retrieving and alert generation. For example, planned lead times from ERP are used to plan interventions instead of actual lead times, which may deviate from planned. Alerts are often generated based on ad-hoc rules, whose performances are not (fully) assessed. The number of alerts generated by the SCT and other tools at A, B, C and D is very high, e.g., 300 to 2000 alerts per week. Yet, a small

Table 5
Interventions at *the five* (priority score 1 for not used and 3 for extensively used).

Interventions and actions		OEMs				
		A	B	C	D	E
Stock reallocation	Proactive stock reallocation from upstream	3	1	2	2	1
	Reallocation of returned parts	2	2	3	1	1
	Disposing parts	2	2	3	1	1
	Skipping regular replenishment	1	1	2	2	1
	Dynamic inventory rationing	3	2	3	1	1
	Backorder clearing	3	2	3	1	1
	Reallocation of a reserved part for preventive maintenance	1	2	3	2	3
Expediting	Emergency shipment from upstream (reactive)	3	3	3	3	3
	Emergency shipment from upstream (proactive)	1	2	3	1	1
	Lateral transshipment (reactive)	3	3	3	1	1
	Lateral transshipment (proactive)	1	2	2	1	1
	Expediting parts in repair	2	3	3	2	2
	Priority scheduling of repair jobs	2	3	3	2	1
	Expediting a new buy	3	3	3	2	2
	Overtime planning of repair personnel	1	1	1	2	1
Cannibalization	Removing a part from another nearby system	1	2	3	1	2
Dynamic capacity allocation, lot sizing	Repair capacity allocation	1	1	2	1	1
	Lot sizing ready-to-repair parts	2	2	1	2	3
Joint ordering	Consolidating shipments	2	2	2	2	3
No intervention	Waiting for arrival of a part from the pipeline	3	3	3	2	3
	Fulfillment of part request from an external source (lost sales)	2	2	3	2	1

Table 6
Information use and control at *the five* (1 for no use, 3 for extensive).

Supply and Demand Information Use and Control	OEMs				
	A	B	C	D	E
Getting status update about the external supply (your suppliers') process	3	3	3	2	2
Having control over external supply process	2	2	2	1	1
Having prediction models/demand forecasts	3	3	3	2	3
Using prediction models/forecast in planning operations interventions	3	2	3	1	3

portion (15% to 50%) of alerts is identified as high priority. This brings the necessity of prioritizing alerts and attention on alert generation.
- Alert generation is based on traditional forecasting techniques, e.g., moving average and simple exponential smoothing. During direct discussions, *A* informed us that short-term forecasts based on real-time information is actually what practice needs. One possibility, which we are aware of at some *others,* is to use condition monitoring of installed systems.

4.2.4. Information content

We observe that information content is limited by information use and control. Table 6 summarizes the information use and control at *the five*. This leads to the following observations:

- *The five* all have information about their external supply process. However, they have limited or no control over their supply process.
- Four of *the five* use demand forecasts for operational interventions. This is less relevant for *D* since its customers control their own inventory and they do not share detailed demand information, which is typical in defense industry. Therefore, *D* observes replenishments orders by its customers and not the actual part usage.

4.2.5. Relation between tactical and operational planning

Table 7 summarizes the approaches that define the link between tactical and operational planning at *the five*. Only *A* and *B* use an integrated approach to update tactical planning when it deviates. *B* uses a rolling horizon approach and updates tactical planning automatically, whereas *A* often uses simple manual adjustments. Others ignore this link or use a hierarchical approach, therefore, tactical planning may often be overruled by operational planning at these OEMs.

4.2.6. Other model-driven elements

Table 8 summarizes *the five* with respect to other model-driven elements. The following observations are common (except for *E* which does not operate an SCT):

- Deterministic demand and supply models are widely used. The uncertainty of demand and supply, which is crucial in estimating downtime risk, is often neglected.
- Operational planning problems are solved by using either simple rules (ad-hoc thresholds) or made manually based on expert knowledge.
- Using a finite horizon model is quite common in operational planning.

5. Overview of papers on operational planning of spare parts

First, we explain in Section 5.1 how we selected relevant papers. Next, in Section 5.2, we summarize the papers according to our framework, and develop some observations and insights.

5.1. Selection of papers

Corresponding the scope of our review, we include papers that consider operational interventions in spare parts planning as presented in Section 3.1.1. In case of doubt, we decide to include the paper, unless it would lead to including an excessive amount of papers. Therefore, we make the following inclusion decisions:

- We include the papers that are directly related to spare parts, illustrating a numerical example considering spare parts, or motivated by spare parts applications. We exclude papers on generic models and methods that are likely to be applicable to operational spare parts inventory and supply planning, e.g., the vast literature on priority scheduling in production-inventory systems (Zheng & Zipkin, 1990; Zipkin, 1995). That area may require a separate review in itself.

Table 7
Approaches to define the relation between tactical and operational planning.

Approaches	OEMs				
	A	B	C	D	E
No link, Hierarchical, Integrated	Integrated	Integrated	Between no link -hierarchical	No link - Hierarchical – Integrated (depending on service contracts)	NA hence No link

Table 8
Comparison of the five with respect to other model-driven elements.

Elements	OEMs				
	A	B	C	D	E
Optimization methods for decision problems	Simple decision rules	Simple decision rules	Simple decision rules	Manual/Simple decision rules (depending on service contracts)	NA
Demand and supply model	Mix of deterministic and stochastic	Mix of deterministic and stochastic	Mix of deterministic and stochastic	Deterministic	NA
Planning horizon	Finite	Finite	Finite	Infinite/Finite (depending on service contracts)	NA

- We include papers focusing on operational planning or addressing it jointly with tactical planning. We exclude papers that focus on the impact of fixed operational interventions on tactical parameters (e.g., impact of using emergency shipments on base stock levels).

Considering these points, we identified more than 50 papers. Table A.1 in Appendix A classifies the papers according to the practice-driven and the model-driven elements of our framework. In the last column, we also clarify the relevance of the papers to spare parts planning. We found most papers under the topics of operational planning and dynamic models for spare part supply chains. We also explored related fields like servitization and (performance based) contracting for the spare part business. In the latter fields, we only found conceptual work and quantitative models focusing on tactical decisions.

5.2. Observations and insights

In this section, we summarize the papers reviewed and provide some observations and insights according to our framework.

5.2.1. Key performance indicators

The majority of the papers reviewed assumes a cost-oriented model, including shortage costs, rather than a service-oriented model. Shortage costs are often not directly observable in practice, but they are closely related to service measures (Van Houtum & Zijm, 2000). There exist two types of shortage costs: backorder costs per unit short per unit time (Abouee-Mehrizi, Baron, & Berman, 2014; Axsäter, 2003; Howard, Marklund, Tan, & Reijnen, 2015) and lost sales penalty costs per unit (Evers, 2001; Minner, Silver, & Robb, 2003; Van Wijk, Adan, & van Houtum, 2013). Service-oriented models, consider as KPIs the probability of no stockout (Alfieri, Pastore, & Zotteri, 2017; Johnson & Scudder, 1999; Lee, Jung, & Jeon, 2007), number of machines down or the system availability (Fisher, 1989; Sherbrooke, 2004), and expected delay in repair shops (Hausman & Scudder, 1982). Fill rate and time-based fill rate which are the most commonly used KPIs by *the five* are considered in a few papers only (Kukreja & Schmidt, 2005; Simao & Powell, 2009; Yang, Dekker, Gabor, & Axsäter, 2013).

In multi-item settings, aggregation on parts corresponds to the so-called system approach. Among the papers reviewed, the system approach (i.e., no segregation) is widely used in priority scheduling of jobs in repair shops (Chua, Scudder, & Hill, 1993; Johnson & Scudder, 1999; Liang, Balcioglu, & Svaluto, 2013).

Customer segregation is used when multiple customer classes of varying importance exist. In operational planning, differentiation can be performed using dynamic inventory rationing: Should we satisfy non-premium demand now, or reserve stock to satisfy future premium demand? See Evans (1968); Topkis (1968), and Kaplan (1969).

5.2.2. Interventions

The current literature is rich in stock reallocation and expediting. Many papers investigate a single intervention such as reallocation of after-repair stock in a two-echelon system (Miller, 1974; Rezaei Somarin, Chen, Asian, & Wang, 2017), dynamic inventory rationing (Teunter & Klein Haneveld 2008; Fadıloğlu & Bulut, 2010; Hung and Hsiao, 2013), lateral transshipment (Axsäter, 2003; Minner et al., 2003), emergency shipment from upstream (Janakiraman, Seshadri, & Sheopuri, 2014; Johansen & Thorstenson, 2014), and expediting a repair process (Ormon & Cassady, 2004; Verrijdt, Adan, & de Kok, 1998). We note that all papers about expediting study expediting a replenishment or a repair process that has not started yet.

Many papers combine different interventions. For example, cannibalization is often studied together with priority scheduling (Fisher & Brennan 1986; Salman, Cassady, Pohl, & Ormon, 2007). Stock reallocation and expediting are usually combined (Hillestad, 1982; Howard et al., 2015; Pyke, 1990). Others combinations include various types of expediting interventions such as priority scheduling of repair jobs and overtime planning of repair personnel (Scudder, 1985; Scudder & Chua, 1987), capacity allocation (Caggiano, Muckstadt, & Rappold, 2006; Frazzon, Israel, Albrecht, Pereira, & Hellingrath, 2014) and lot sizing (Chua et al., 1993; Johnson & Scudder, 1999).

When combined, certain interventions appear to have complementary benefits while others substitute each other. For example, Sherbrooke (2004) shows that the benefit of proactive stock reallocation is less significant under reactive lateral transshipments. Proactive stock reallocation has more impact when a repair shop uses priority scheduling. That is, reactive lateral transshipments substitute proactive stock reallocation, whereas priority scheduling complements the benefits. Whether interventions are complementary or substitutable also depends on the environment. Fisher (1989) shows that priority scheduling in a repair shop result in a lower expected number of machines down when combined with cannibalization, demonstrating that the two interventions are complementary. In contrast, priority scheduling does not significantly influence the service level when cannibalization is allowed in Salman et al. (2007). They consider relatively short cannibalization times. As a result, cannibalization substitutes priority scheduling.

Interventions considered in the literature are -almost equally- identified as proactive or reactive and as event-based or

time-based. Although reactive interventions are identified as event-based and proactive interventions are time-based in many instances, there are a few exceptions. Glazebrook, Paterson, Rauscher, and Archibald (2015) and Lee et al. (2007) include reactive lateral transshipments in a periodic review setting, where demand fulfillment takes place at the end of the period. Paterson, Teunter, and Glazebrook (2012) consider reactive lateral shipments as opportunities for proactive lateral transshipments in a continuous review setting (where demand fulfillment takes place in real time).

Several papers investigate the value of integrating reactive and proactive interventions. The literature focuses on proactive lateral transshipments (Lee et al., 2007; Paterson et al., 2012; Seidscher & Minner, 2013), proactive shipments from upstream (Grahovac & Chakravarty, 2001), and overtime planning of repair personnel (Scudder, 1985; Scudder & Chua, 1987). Main findings are: Proactive interventions can lead to significant cost reductions and service level improvement in comparison to using reactive interventions only (Seidscher & Minner, 2013). Proactive lateral transshipments are shown to be more effective when transportation cost is low (Lee et al., 2007). Overtime planning of repair personnel is shown to be most effective when repair times are relatively short (Scudder, 1985) and when the arrival of failed parts are predictable (Scudder & Chua, 1987). With proactive shipments from upstream, inventories are not always reduced, but savings on total cost are obtained by reallocating inventory from upstream to downstream (Grahovac & Chakravarty, 2001). Papers on how to use advance demand information, e.g., age and condition of installed systems, focus only on tactical planning. These papers do not consider the operational interventions discussed in Section 3.1.2 (Elwany & Gebraeel, 2008; Topan, Tan, van Houtum, & Dekker, 2018).

5.2.3. Triggers for interventions

We identify four streams of research that are related to alert generation in operational planning: (i) supply chain disruption and risk management (Ho, Zheng, Yıldız, & Talluri, 2015), (ii) spare parts demand forecasting (Altay & Litteral, 2011; Bacchetti & Saccani, 2012; Eaves & Kingsman, 2004; Syntetos, Babai, & Altay, 2012), (iii) condition monitoring in spare parts planning (Lin, Basten, Kranenburg, & van Houtum, 2017), (iv) supply chain event management (e.g., Ijioui, Emmerich, & Ceyp, 2008). Streams (i)-(iii) mainly focus on long-term (tactical) planning. In contrast, stream (iv) focuses on supply chain monitoring using real-time information, detecting realized or potential deviations from plans, and intervening when needed (Otto, 2003) or using statistical process control to generate alerts (Pfohl, Cullmann, & Stolzle, 1999). To our best knowledge, the only papers related to spare parts in this stream are Braader and Montanus (2008), and Bausmann, Schmidt, and Bauer (2008), with an application in aviation. Yet, the two papers do not address how companies identify trends, recognize possible deviation from plans, generate alerts, and determine interventions. Therefore, we exclude these papers from Table A.1.

5.2.4. Content of information

Commonly used real-time information includes net stock level, total pipeline stock level, status of pipeline stock, and number of parts waiting for repair. Other information types are problem specific, e.g., remaining time until next review (Kaplan, 1969; Lee et al., 2007), remaining time until the end of planning horizon (Topkis, 1968), the time an order spent in a supply stage (order progress, Gaukler, Özer, & Hausman, 2008), and customer service contract termination date (Pinçe, Frenk, & Dekker, 2015), obsolescence and/or relocation of spare parts (Pinçe & Dekker, 2011), age and/or condition (degradation) of installed units (Elwany & Gebraeel, 2008), number and status of demand signals (Topan et al., 2018).

System performance can be improved significantly by increasing the level of information. For example, the use of status of pipeline stock information is considered for lateral transshipments (Minner et al., 2003; Yang et al., 2013) and emergency shipments from upstream (Gaukler et al., 2008; Howard et al., 2015). In case of long and uncertain replenishment times, high demand rates, and high intervention costs, these information is highly beneficial for decision making. Moreover, for systems with nonstationary demand, demand behavior might depend on the so-called "states of the world". Compared to ignoring the non-stationarity in demand, states of the world information can lead to significant cost reductions (Arts, Basten, & van Houtum, 2016). Finally, the use of real-time inventory information can result in significant inventory reduction in a repair shop when priority scheduling is possible (Hausman & Scudder, 1982).

5.2.5. Relation between operational and tactical decisions

Most papers assume a hierarchical relation between operational and tactical planning where tactical parameters (inventory policy, repair capacity) are considered as *fixed* (Caggiano et al., 2006; Chua et al., 1993). Some use a *sequential approach*, where they first optimize tactical decisions and next operational decisions (Axsäter, 2003 and Kukreja & Schmidt, 2005; Çömez, Stecke, & Çakanyıldırım, 2012; Fleischmann, van Nunen, & Gräve, 2003; Rezaei Somarin et al., 2017). One third of the reviewed papers consider an *integrated approach*, i.e., solving tactical and operational planning *simultaneously* or *iteratively* by using a feedback loop to facilitate two-way interaction between operational and tactical planning. Only two papers consider an iterative approach (Frazzon et al., 2014; Schneeweiss & Schröder, 1992). In the papers that consider integrated approach, operational decisions include expediting (Glazebrook et al., 2015; Grahovac & Chakravarty, 2001; Ormon & Cassady, 2004; Salman et al., 2007), tactical decisions typically involve finding base stock levels. Hoadley and Heyman (1977) is the only paper that do not state a clear link between tactical and operational decisions since they study a single period problem.

Although integrated approaches outperform sequential approaches in cost reduction and service level improvement, computational complexity is high. Papers focus on a single tactical and/or operational decision to ensure tractability. The integrated approach is not investigated for practical problems with multiple interventions and complex supply chain structures. Papers on how to use advance demand information, age and condition of installed systems focus on tactical planning. The relation between tactical and operational planning is not applicable for these papers.

5.2.6. Demand and supply characteristics

Demand for spare parts is often characterized by Poisson or compound Poisson processes (Abouee-Mehrizi et al., 2014; Axsäter, 2003), and sometimes by a general (discrete) probability distribution (Caggiano et al., 2006; Gaukler et al., 2008).

Regular replenishment lead times are typically assumed to be deterministic. For repair shops, it is common to assume exponential repair lead times (Liang et al., 2013; Tiemessen & van Houtum, 2013) or a single server queueing system with constant processing times (Hausman & Scudder, 1982; Scudder, 1984). A few papers consider phased-type distributions (Arts et al., 2016; Gaukler et al., 2008). Indeed, the accuracy of the model for slow moving items is not very sensitive to the shape of the lead time distribution if a stockout is followed by an emergency supply (Verrijdt et al., 1998). This finding is closely related to the insensitivity of the M/G/c loss system for the shape of the service time distribution. We also note that zero lead time is a widely used assumption for interventions such as emergency shipments from upstream or lateral transshipments (Evers, 2001; Glazebrook et al., 2015).

We identify two approaches in modelling ADI: The first approach uses a non-Markovian stochastic process (e.g., random coefficient, Wiener process) to formulate degradation of installed systems. Using this, the papers estimate remaining useful life, plan preventive part replacements, and order the parts needed (Elwany & Gebreel, 2008; Louit, Pascual, Banjevic, & Jardine, 2011 Li and Ryan, 2011). The second approach models the degradation process as a Markov Chain. This is then used to predict future ADI, replenishment decision is formulated as a Markov Decision Process (Lin et al., 2017; Topan et al., 2018). Topan et al. (2018) differ from others by considering three imperfectness measures of ADI, i.e., false positives, false negatives and the prediction interval. Deshpande, Iyer & Cho (2006) differ from these two main streams since in this paper the spare parts that are older than a certain threshold age are considered as advanced demand orders and they immediately trigger replenishment orders.

5.2.7. Planning horizon

More than half of the papers conduct an infinite horizon or steady state analysis (Fadıloğlu & Bulut, 2010; Grahovac & Chakravarty, 2001). Infinite horizon models provide analytical results, closed-form solutions, and/or insights about the optimal policy structures. These models can be considered as an approximation for short-term operational decisions. Yet, their accuracy needs to be tested. On the other hand, a finite horizon is relevant for short-term operational decisions. Yet, deriving analytical results is difficult. Therefore, evaluation and analysis over a finite horizon is conducted mainly by simulation. Some of these models consider specific characteristics due to a finite horizon. For example, Caggiano et al. (2006) formulate the problem of defining repair capacity and stock allocation in a two-echelon repair and distribution system. They solve the finite horizon problem in a rolling horizon manner. Pinçe et al. (2015) develop inventory control policies that consider obsolescence due to contract expirations. The idea is to reduce base stock levels by skipping regular replenishments before obsolescence (i.e., the end of horizon). They apply numerical inversion of generating functions to do the calculations.

5.2.8. Supply chain structure

There are supply chain settings that are widely used for certain interventions. These are:

- A two-echelon inventory system with multiple downstream locations for proactive or after-repair stock reallocation and emergency shipment from upstream (Büyükkurt & Parlar, 1993; Caggiano et al., 2006; Howard et al., 2015; Rezaei Somarin et al., 2017).
- A single echelon system with multiple locations for lateral transshipment (Minner et al., 2003; Paterson et al., 2012; Seidscher & Minner, 2013).
- A single item single location for dynamic inventory rationing (Chew, Lee, & Liu, 2013; Hung, Chew, Lee, & Liu, 2012; Kaplan, 1969).
- A multi-item single repair shop for priority scheduling of repairs (Chua et al., 1993; Johnson & Scudder, 1999). Few priority scheduling papers consider multi-indenture systems (Abell, Miller, Neumann, & Payne, 1992; Sherbrooke, 2004) and/or multiple machines/work centers (Hausman & Scudder, 1982; Schneeweiss & Schröder, 1992).

5.2.9. Optimization and evaluation methods

Several papers formulate problems using a *Markov decision process* (MDP), investigate optimal policies. These papers typically assume an infinite planning horizon. As such, they can perform a steady state analysis, characterize the optimal policies, and generate some generic insights. For example, Topkis (1968) shows that the optimal inventory rationing policy is dynamic. The author investigates the conditions under which the optimal rationing policy between two replenishment instants is determined by a set of threshold levels. Under this policy, demand of a given class is satisfied at a given time only if no demand of a more important class remains unsatisfied and the stock level does not fall below the threshold level for that class. Evans (1968) and Kaplan (1969) derive similar results for two demand classes considering lost sales and backordering, respectively. Arts et al. (2016) investigate the optimal expediting policy and provide closed-form necessary and sufficient conditions that determine whether the optimal policy is a threshold-type policy or a no-expediting policy.

Although many real-life operational problems could be formulated as an MDP, they cannot be solved due to the curse of dimensionality (e.g., when there exist multiple customers, locations, and items). Some papers maintain tractability by proposing *heuristic policies/decision rules* and use *simulation* for performance evaluation. In what follows, we exemplify heuristic policies and decision rules proposed for after-repair stock reallocation, dynamic inventory rationing, emergency shipment from upstream, expediting repair, and cannibalization.

For after-repair stock reallocation, simple *decision rules* are proposed. For example, the repaired part can be sent to the base: (i) with the minimum inventory position, (ii) the base from which the part originated, and (iii) the base with a backorder for the longest time, i.e., first-come-first-served, FCFS (Büyükkurt & Parlar, 1993; Miller, 1974). The simulation study of Büyükkurt and Parlar (1993) shows that the first policy performs best according to the expected number of backorders criterion, but performs worst according to the expected downtime per base criterion (i.e., percentage of time the operation at a base is shut down due to the availability of less than the minimum required number of machines). Abouee-Mehrizi et al. (2014) propose a heuristic policy which allows a location to be prioritized when the inventory level hits a negative threshold. The proposed heuristic policy outperforms the FCFS policy in terms of total cost.

For dynamic inventory rationing, heuristics employ a continuous time approximation (Teunter & Klein Haneveld, 2008), continuous replenishment flow assumption (Fadıloğlu & Bulut, 2010), or myopic policies (Hung et al., 2012). In general, these methods work well compared to static threshold-type policies. For the dynamic demand fulfillment problem, Tiemessen, Fleischmann, van Houtum, van Nunen, and Pratsini (2013) develop an allocation policy that minimizes the sum of direct cost and an approximation of future costs (one-step look-ahead policy). Numerical experiments inspired from IBM's spare parts network in Europe show that the proposed allocation policy provides significant cost savings compared to a simple but widely used static allocation policy that ignores customer differentiation.

For emergency shipments from upstream, Janakiraman et al. (2014) investigate the performance of the so-called tailored base-surge (TBS) policy. Under this policy, a base level of demand is met by regular replenishment, and demand surges are met by emergency shipments. They show that the cost difference between the best TBS policy and the optimal policy decreases dramatically as the lead time of regular replenishment increases.

For expediting repair activities by priority scheduling, Tiemessen and van Houtum (2013) evaluate various dynamic scheduling policies and show that the myopic rule outperforms the other rules in terms of aggregate downtime reduction. For a capacitated repair shop, Liang et al. (2013) also propose a myopic policy, which is easily implementable and yields near-optimal solutions.

For cannibalization, the performance of heuristic policies are highly dependent on the problem environment (i.e., repair lead times, demand distribution, manpower constraints, and spare part

stock levels). Fisher and Brennan (1986) and Fisher (1989) develop simulation models to assess the performance of different policies. The best policies found are those that allow cannibalization when it does not delay other maintenance jobs. The policy of complete cannibalization is shown to perform poorly, except when the utilization rate of servers is low and cannibalization time is short. The simulation results of Fisher (1989) show that a dynamic cannibalization policy (that considers real-time information on the number of machines down and the availability of servers) performs best and significantly better than no cannibalization.

Other solutions methods used for dealing with the computational challenges in real-life problems include *linear programming* and *approximate dynamic programming*. Caggiano et al. (2006) formulate repair capacity and stock allocation problem using a large scale linear program (and run this model in a rolling horizon manner). Frazzon et al. (2014) study proactive stock reallocation and lot sizing decisions in a spare parts production and distribution system. They formulate a mixed integer linear program (MILP), integrating the capacitated lot sizing problem (production decisions) and the network flow problem (distribution decisions). Simao and Powell (2009) propose an approximate dynamic programming approach to determine proactive stock reallocation decisions. Although their model and solution strategy are not compared with other methods, they have been tested at an aircraft manufacturer and results are found to be promising.

6. Conclusions and open issues

Based on discussions with the five companies and our previous experiences with other companies, and after exploring the papers in the literature, we draw conclusions and identify future research directions for operational planning of spare parts and developing SCTs. We present our findings by associating them with the elements of our framework (this does not correspond to an order of priority). We also summarize the most promising research directions.

6.1. Key performance indicators

Service providers may face penalties if target service levels over a fixed period (quarter, year) are not met as specified in *service level agreements* (SLAs). We did not find papers incorporating real-time service performance in operational decision making.

Customer differentiation is mainly used in tactical planning by *the five*. Research papers show that dynamic rationing can bring significant benefits. In addition, priority scheduling and emergency shipments can be used based on the real-time service performance for each customer. How to use these interventions in operational management of SLAs and its potential benefits need further investigation. This has not been investigated in the literature despite its practical value.

System approach is common in tactical planning, but companies still focus on individual item performance in operational planning. As SLAs are often on system level, how to implement system approach at operational level may bring benefits to practice and contribute to the literature.

6.2. Interventions

In SCTs, exception messages are automatically generated. The corresponding interventions are manual, they are not fully assessed and they do not benefit from decision rules proposed in the literature. Several decision rules exist in the literature for operational interventions such as priority scheduling, after-repair stock allocation, and cannibalization. Such rules can be used for *automated processing of alerts* in practice. Yet, their effectiveness may depend on the situation. Full automation may be feasible for cheap fast movers, as the number of observations will be high and the cost impact will be low. The reverse is true for expensive slow movers. These decisions may be left to planners with some support from planning tools.

Research papers show that certain interventions have complementary benefits (e.g., priority scheduling and stock reallocation), while others substitute each other (e.g., lateral transshipments and stock reallocation). Identifying the conditions which make an intervention *complementary/substitutable* and quantifying its consequence would help decision makers to select the best intervention.

Research papers focus on expediting replenishment/repair processes that have not started yet, but not on *expediting ongoing processes*. The latter makes sense in practice and typically requires informal processes, e.g., calling a supplier to request early delivery or repair expediting.

Parts reserved for preventive maintenance can be used as an additional supply source in order to prevent stock-outs or stock-out risk. This leads to postponing planned preventive maintenance, bearing an increased failure risk for the associated system. To our knowledge, this intervention has not been investigated in the literature.

6.3. Triggers for interventions and alert generation

In SCTs, alerts are generated when projected stock levels deviate from pre-defined tactical plans. Sometimes, arbitrary threshold values have been used. *Finding the optimal thresholds* contributes to practice. Alert systems generate too many exception messages. *Prioritizing exception messages* allows planners to focus on the most important problems.

6.4. Content of information

Many papers assume negligible lead times for emergency shipments and lateral transshipments and constant lead times for regular replenishments. One way to deal with supply uncertainty is to use real-time *advance supply information* in planning, e.g., status updates of expected delivery times. To our best knowledge, only one paper focuses on the use of supply information in the operational planning of spare parts (Gaukler et al., 2008), although there are papers in other applications (Kouvelis & Tang, 2012; Mogre, Wong, & Lalwani, 2014). Considering long and uncertain external supply lead times, using advance supply information to plan operational interventions proactively could make an important contribution to practice.

Traditional forecasting methods for spare parts is based on historical information (Altay & Litteral, 2011; Bacchetti & Saccani, 2012; Eaves & Kingsman, 2004; Syntetos et al., 2012). Yet, there are issues about short-term forecasting and its use in operational spare parts planning, e.g., how to collect information, how to make short-term predictions, and considering that this information may be imperfect, how to use it in planning. Companies do not use *advance demand information* (ADI), e.g., age and/or condition information of installed bases, or *advance supply information* (ASI), e.g., status updates of expected delivery times. Operational planning may benefit from such information to predict deviations in pans in the short run. A number of studies focus on the use of ADI in inventory planning (Topan et al., 2018). Nevertheless, these papers focus on tactical planning and study the use of ADI in planning regular replenishments. Current literature on using installed base information focuses on forecasting spare part demand during the end-of-life stage of a product. Using installed base information at operational level across the life cycle of a product is of interest. Literature on ASI is also scarce.

6.5. Relation between operational and tactical decisions

Many papers investigate the impact of operational interventions under fixed tactical planning parameters (e.g., target inventory levels) that are decided arbitrarily. However, this impact may be strongly influenced by the quality of tactical planning. A good tactical planning should consider the flexibility offered by operational interventions, e.g., emergency shipments imply shortened lead times. The link between tactical and operational planning is crucial. Yet, integrating tactical and operational planning is not trivial and require structured methods. Despite several papers, a framework and a unifying approach to integrate operational and tactical spare parts planning is lacking in the literature. Frameworks on spare parts planning (Driessen et al., 2015; Hu et al., 2018) describe the operational and tactical planning in practice, however, they do not focus on methods integrating them. Practice-driven approaches such as rolling horizon, feedback loops may facilitate this integration. The literature on spare parts planning is surprisingly scarce in using such methods to *integrate operational and tactical planning*.

6.6. Optimization and evaluation methods

Operational interventions are quite diverse and often interlinked. Selecting the right intervention depends on complex real-time information about expected repair time, spare parts' inventory position, and number of machines down. These complex systems and complex decision problems are generally modelled by using simulation, and the proposed decision rules are assessed in simulation experiments. Yet, these simulation models are often system specific and include a limited number of interventions (usually two or three). Similarly, existing analytical models investigate either one or a small set of these interventions. Few papers consider more than two interventions. A *unified and holistic approach* should be able to solve large scale problems with typically thousands of spare parts and tens of locations. Approximate dynamic programming may cope with this complexity (Powell, 2011).

Based on our discussions with practitioners and our overall literature review, we conclude that the most promising research directions are as follows: (i) incorporating system level SLAs with fixed time windows into operational planning, (ii) managing and automated processing of alerts, (iii) integrating tactical and operational plans, (iv) using advance supply and demand information to plan operational interventions proactively, and (v) developing a unified and holistic approach for selecting interventions in large scale systems. Furthermore, we exclude some topics related to operational spare parts planning from our review. Among these topics, scheduling at repair shops, service technician planning, other service control layers such as data perception, data storage and control, data application are possible topics of a future review.

With this review, we hope to contribute to (i) a better understanding of current practice in operational planning of spare parts planning and the need for SCTs to support these decisions, (ii) identifying the methods and approaches available in scientific papers that practitioners need and (iii) highlighting the issues that deserve further investigation to stimulate the future research.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2019.03.026.

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